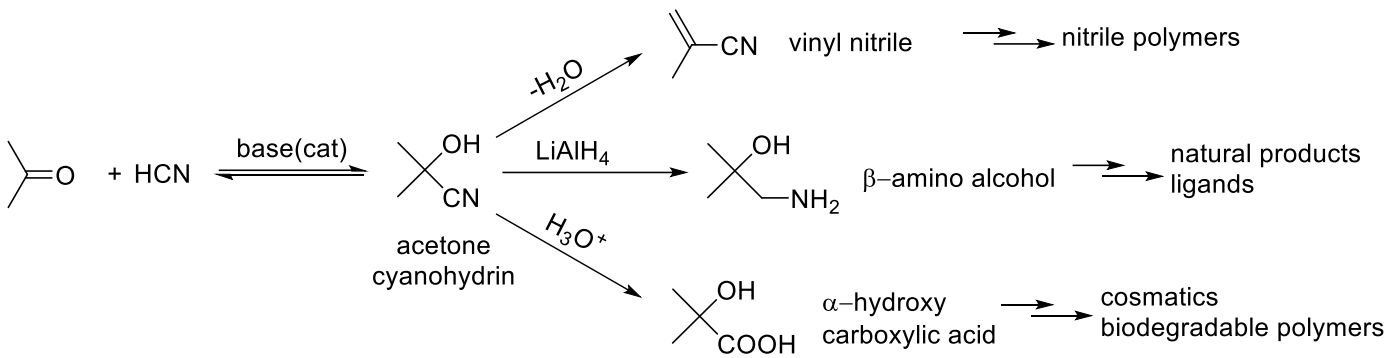
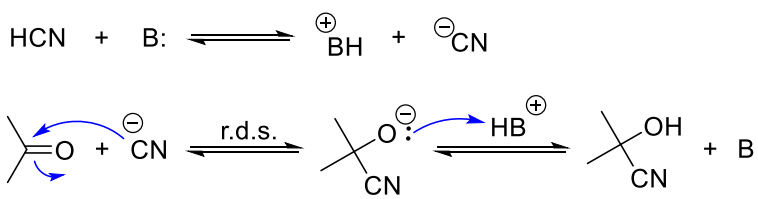
Reactions with C – nucleophiles

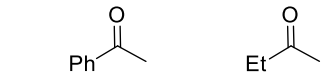
Cyanohydrin formation:



Mechanism: (Base catalyzed)

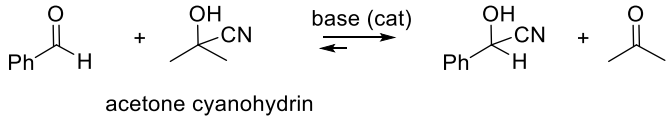


- Nucleophilic attack of $^{-}$CN to C=O is r.d.s. Rate = k[ketone] [$^{-}$CN]
- The reaction is reversible. The position of the equilibrium depends upon electrophilicity and steric.

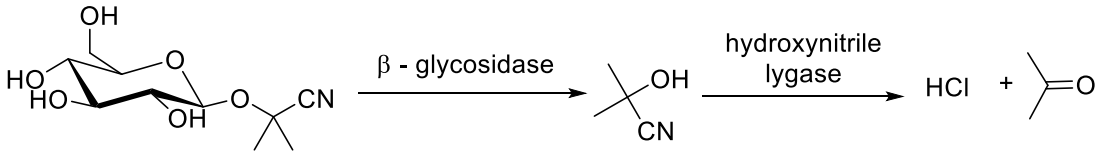


K 212 28

- HCN is toxic gas. Alternative reagents TMSCN, $Me_2(OH)CN$.



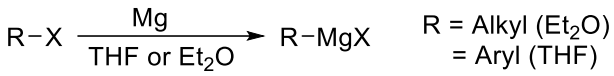
- Food toxicity in cassava



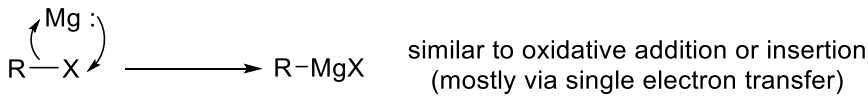
- Raw cassava is toxic (~ 50 mg of HCN/100 g of cassava)
- Fermented: raw cassava is crushed with water, then fermented (to release HCN) then washed and cooked.
- Some acute toxicity still exists.

Reactions with RMgX and RLi: Grignard reagents: RMgX

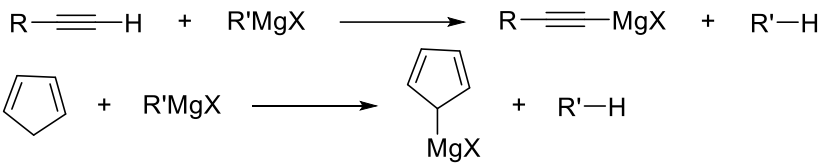
Preparation:



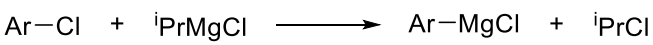
Mechanism: (approximately)



- va deprotonation of acidic C-H bonds.

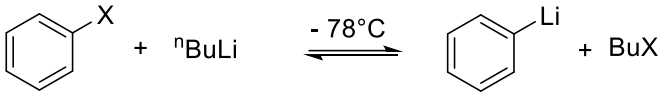


- via halogen exchange.



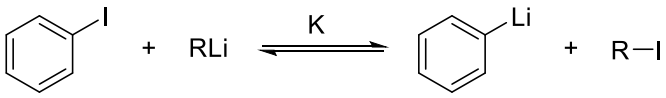
- **Organolithium compounds:**

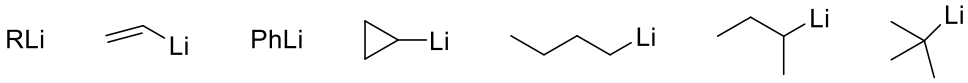
Preparations: via lithium exchange reaction (most common)



rates : X = I >> Me_3Sn > Br > Cl

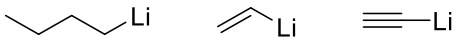
Thermodynamics





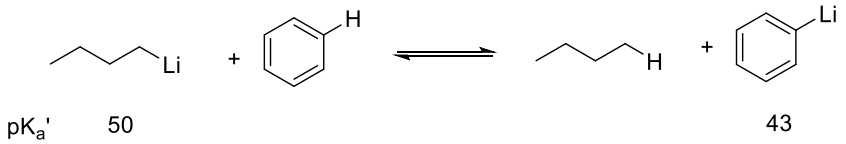
K	0.004	1	10	7500	10^7	$>>10^7$

- Can be explained by considering the acidity of C-H bonds



pK_a	50	41	24

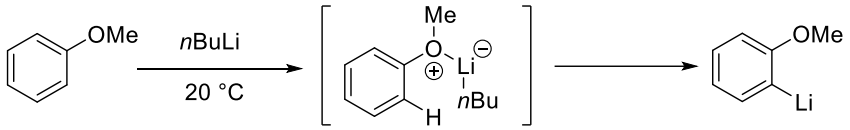
Thermodynamics vs Kinetics



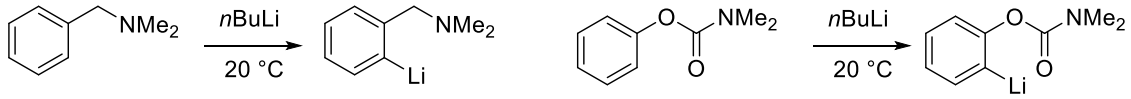
⇨ thermodynamically K = $10^{50-43} = 10^{7}$

⇨ but the reaction is kinetically very slow

Directed ortho-lithiation (kinetic push)

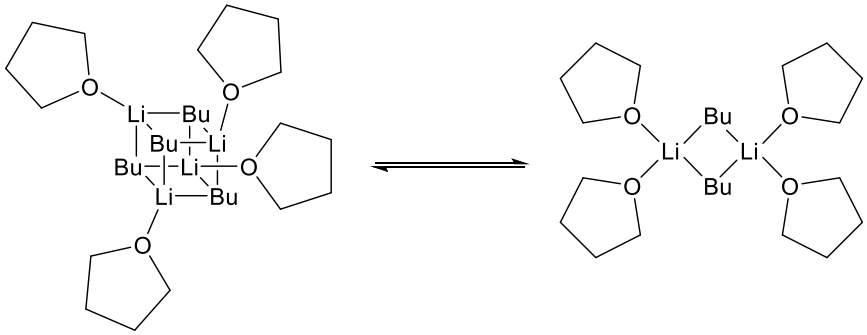


Similarly

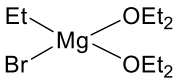


Structure of 'R-M' in solutions

● ^{n}BuLi in THF



● EtMgX in Et_2O



Monomeric

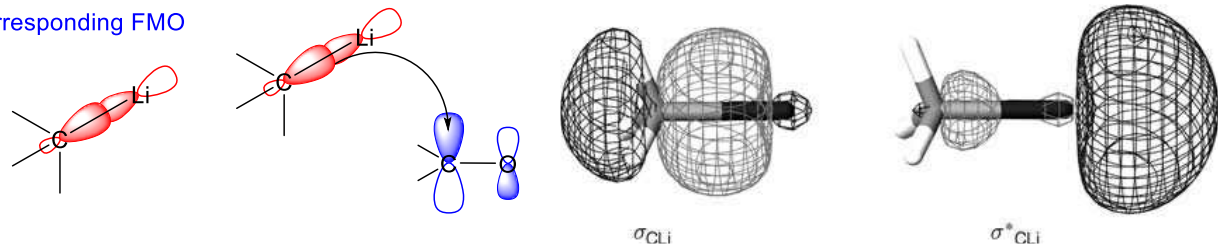
Dimeric possible

● Schlenk equilibrium

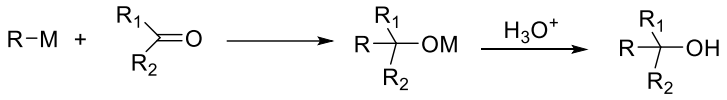
$$RMgX \rightleftharpoons R_2Mg + MgX_2$$

Roughly 'R–M' should be more appropriate description (better NOT to write it as R^- M^+)

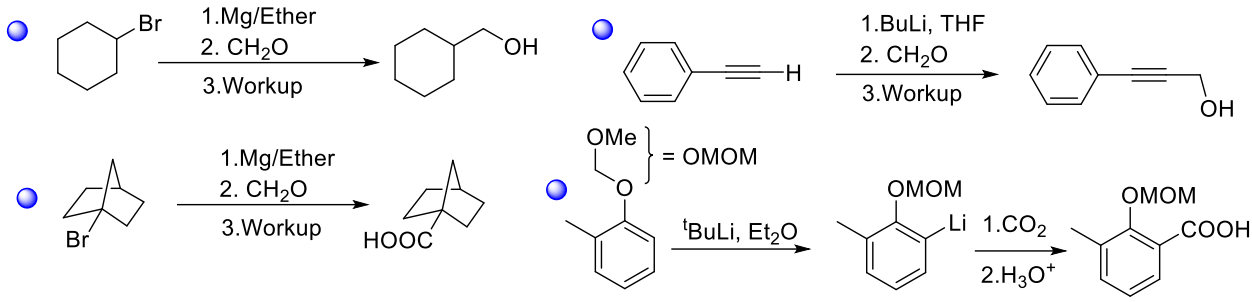
Corresponding FMO



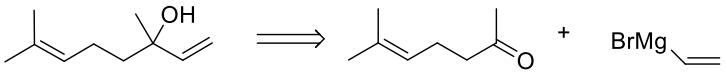
- **Reactions of RM with carbonyl compounds:**



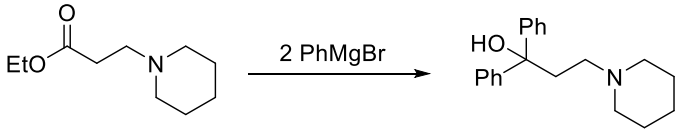
Examples:



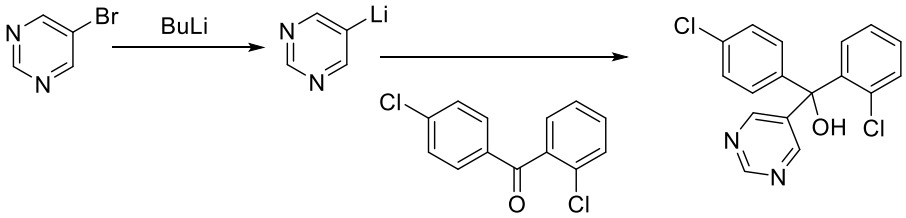
Application:

- 

  Linalool (Perfume compounds)

- 

  fenpiprane (anti-histamine)

- 

  Fenarimol
  (fungicide)